

Portfolio Trading Strategy

Computational Finance

1. Introduction

This project develops a systematic trading strategy using Python and historical data from 20 U.S. stocks. By combining 12-month momentum, EMA-based timing signals, and risk-adjusted portfolio weighting with monthly rebalancing, the strategy aims to balance return and risk in a disciplined way. Backtesting over the 2015–2025 period shows an annualized return of 23.93%, highlighting the potential of data-driven quantitative investing.

2. Data Collection

2.1 Source of Data

The stock data used in this project was sourced from Yahoo Finance, which provides historical price data and basic company information for a wide range of publicly traded firms.

It was selected because it is freely accessible, easy to use, and sufficiently reliable for academic and exploratory financial analysis. All price data was retrieved using Python, allowing the data to be processed and analyzed directly within the same workflow.

2.2 Selection of Criteria

The stocks were selected according to historical performance, market capitalization, and sector diversification. The portfolio combines high-growth firms (e.g., Apple, Microsoft, Amazon, Tesla) with established financial institutions (e.g., JPMorgan Chase - JPM) to improve resilience across market conditions. The tickers and brief profiles of the 20 selected U.S. equities are listed in Table 1.

Stock Symbol	Company Name	Primary Business
AAPL	Apple Inc.	Consumer electronics, computer software, and online services
MSFT	Microsoft Corporation	Software development, computer hardware, cloud computing
AMZN	Amazon.com, Inc.	E-commerce, cloud computing, artificial intelligence
GOOGL	Alphabet Inc. (Google's parent company)	Internet-related services and products, including search, advertising, and cloud computing
META	Meta Platforms (Formerly Facebook)	Social media, virtual reality

TSLA	Tesla, Inc.	Electric vehicles, solar energy generation, and energy storage
BKR	Baker Hughes	Oil services and equipment
JPM	JPMorgan Chase & Co.	Financial services
JNJ	Johnson & Johnson	Pharmaceuticals, medical devices, and consumer goods
V	Visa Inc.	Payment technology
WMT	Walmart Inc.	Retailing
PG	Procter & Gamble Co.	Consumer goods
MA	Mastercard Incorporated	Payment technology
UNH	United Health Group	Health insurance and healthcare services
BAC	Bank of America Corp.	Banking and financial services
HD	The Home Depot, Inc.	Home improvement retailer
NVDA	NVIDIA Corporation	Visual computing technology
ADBE	Adobe Inc.	Software and services focused on creative and document management
PYPL	Paypal Holdings, Inc.	Digital payment
DIS	The Walt Disney Company	Media and entertainment

Table 1. 20 US stocks and profiles

3. Methodology

3.1 Momentum Signal

The momentum signal measures a stock's performance over the previous 12 months using monthly closing prices. The momentum score is defined as:

$$M_i(t) = \frac{P_i(t)}{P_i(t-12)} - 1$$

where $P_i(t)$ is the month-end price of stock i at time t . [1]

3.2 Exponential Moving Averages and Trading Signals

To identify price trends and generate trading signals, this strategy applies Exponential Moving Averages (EMAs). An EMA places more weight on recent price observations, allowing it to respond faster to new market information than a Simple Moving Average (SMA). This makes EMAs more suitable for trend-following systems that require timely reaction to price changes, as noted by Zhou and Jain [4]. The EMA for stock i at time t is defined as:

$$EMA_t = \alpha \cdot P_t + (1 - \alpha) \cdot EMA_{t-1}$$

where:

- P_t is the price at time t ,
- $\alpha = \frac{2}{n+1}$ is the smoothing factor,
- n is the chosen lookback period.

Two EMAs are employed to generate trading signals. A golden cross signals a potential entry, while a death cross indicates a possible exit.

3.3 Risk-Adjusted Scores

Rachev et al. [2] found that momentum strategies based on risk-adjusted performance measures (e.g., Sharpe ratio) provided better risk-adjusted returns than momentum strategies based on total returns in most situations.

To prevent highly volatile stocks from dominating the portfolio, momentum scores are adjusted for risk. Daily returns from the past year are used to estimate annualized volatility:

$$\sigma_i = std(r_i) \times \sqrt{252}$$

where r_i is the daily return of stock i .

The risk-adjusted score for each stock is then computed as:

$$S_i = \frac{M_i}{\sigma_i}$$

where M_i is the momentum score defined earlier.

Stocks with higher momentum and lower volatility receive higher scores, allowing the strategy to favor stable performers rather than purely high-return but unstable stocks. This approach is commonly used in quantitative investing because it balances return potential with risk exposure [3].

3.4 Portfolio Construction Process

3.4.1 Momentum-Based Stock Selection

To construct the portfolio, all stocks in the investment universe are first ranked by their 12-month momentum. For stock (i), the momentum measure is defined as:

$$M_i = \frac{P_i(t)}{P_i(t-12)} - 1$$

Where $P_i(t)$ denotes the monthly adjusted closing price at time (t), and 12 trading months approximates one year. The ten stocks with the highest positive M_i values are selected.

Portfolio weights are obtained by normalizing scores:

$$w_i = \frac{M_i}{\sum_{j=1}^N M_i}$$

Table 2 reports an illustrative momentum score and weights obtained from 30/11/2024–30/11/2025.

Ticker	Momentum Scores	Momentum Weights
GOOGL	0.903474	0.304618
JNJ	0.375743	0.126687
NVDA	0.280656	0.094627
JPM	0.280264	0.094495
TSLA	0.246292	0.083040
WMT	0.206352	0.069574
AAPL	0.180260	0.060777
MSFT	0.170452	0.057470
BKR	0.166624	0.056179
BAC	0.155811	0.052534

Table 2. Momentum Scores, Weights and Top-10 Selection

3.4.2 Risk-Adjusted Weighting

Because stocks with higher volatility tend to exhibit unstable returns, momentum scores are volatility-adjusted to penalize risk. Let σ_i denote the annualized volatility estimated from 252 daily returns:

$$\sigma_i = std(r_i) \times \sqrt{252}$$

where r_i is the daily return of stock i .

The risk-adjusted score is then:

$$S_i = \frac{M_i}{\sigma_i}$$

Portfolio weights are obtained by normalizing scores:

$$w_i = \frac{S_i}{\sum_{j=1}^N S_j}$$

Table 3 provides an empirical example for the top-10 stocks identified in Table 2.

Ticker	Momentum Scores M_i	Annual Volatility σ_i	$S_i = \frac{M_i}{\sigma_i}$	Weights w_i
GOOGL	0.903474	0.331900	2.722132	0.275314
JNJ	0.375743	0.193228	1.944556	0.196671
NVDA	0.280656	0.496417	0.565364	0.057180
JPM	0.280264	0.242495	1.155751	0.116892
TSLA	0.246292	0.652284	0.377583	0.038188
WMT	0.206352	0.251644	0.820014	0.082936
AAPL	0.180260	0.325502	0.553792	0.056010
MSFT	0.170452	0.244070	0.698372	0.070633
BKR	0.166624	0.354336	0.470244	0.047560
BAC	0.155811	0.268845	0.579556	0.058616

Table 3. Example of Volatility Adjustment and Final Weights

3.4.3 Portfolio Return Aggregation

Weights w_i determined at time t are applied to subsequent returns ($r_i(\tau)$)

for $\tau > t$. The portfolio return at time τ is computed as:

$$R_{p(\tau)} = \sum_{i=1}^N w_i \cdot r_i(\tau)$$

Daily portfolio returns are concatenated across rebalancing periods to form the raw strategy series:

$$\{R_p(\tau)\}_{\tau=1}^T$$

This series is then used for performance evaluation (Sharpe ratio, CAGR, drawdown) and for overlaying systematic timing rules to generate the final trading strategy output.

3.4.4 EMAs Signal

Golden and death cross signals are identified using 9-day and 25-day EMAs based on one year of historical data. These parameters are suitable for short-term investors, while longer horizons may use wider EMA windows.[5]

The strategy executes trades at crossover points, and the portfolio performance is shown in Figure 1.

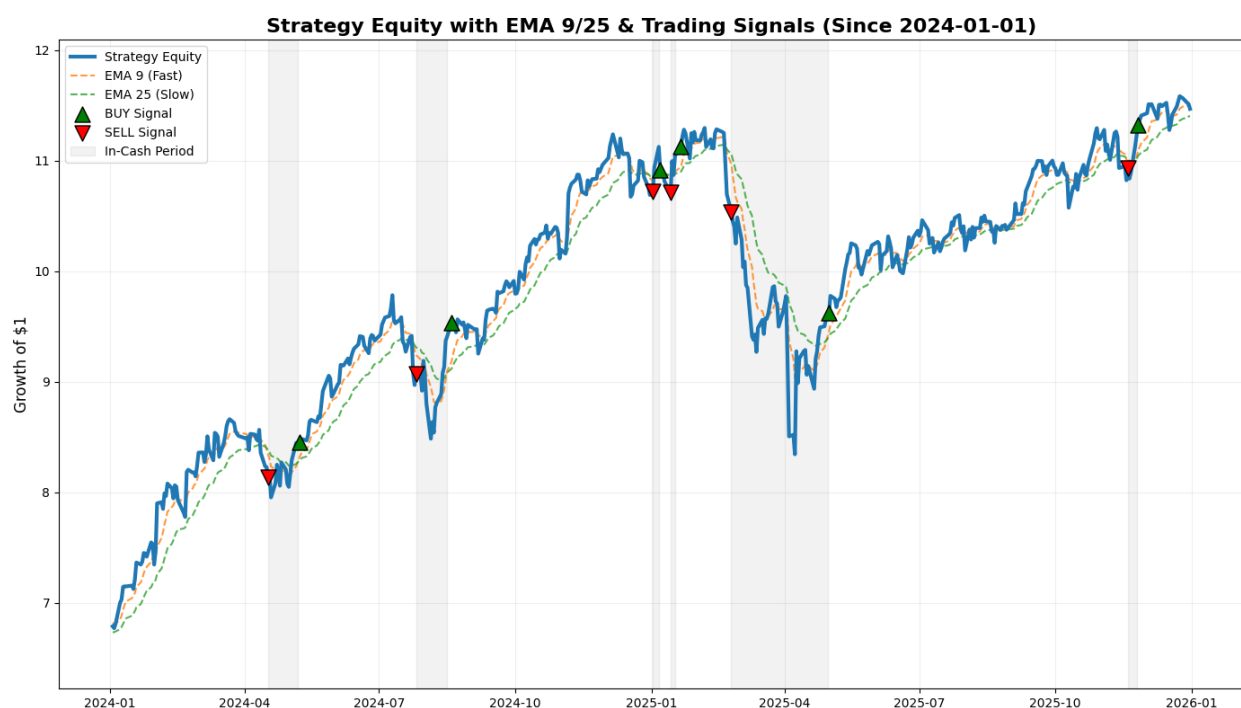


Figure 1. Strategy Equity with EMA and Trading Signal (01/01/2024-31/12/2025)

4. Results

4.1 Backtesting

The strategy was backtested from January 1, 2015, to December 31, 2025. The following section details the empirical findings, comparing the Momentum-Volatility strategy against the S&P 500 benchmark (SPY).

The strategy's performance can be attributed to its ability to capture upward momentum while limiting exposure during major market downturns.

As illustrated in the cumulative return chart (Figure 2), the strategy successfully identified a

golden cross in mid-2020, allowing for full participation in the post-pandemic market recovery.

In contrast, the death cross signals observed in 2022 helped reduce market exposure, resulting in a maximum drawdown of only -22.21%, compared to -33.72% for the benchmark, as reported in Table 4.

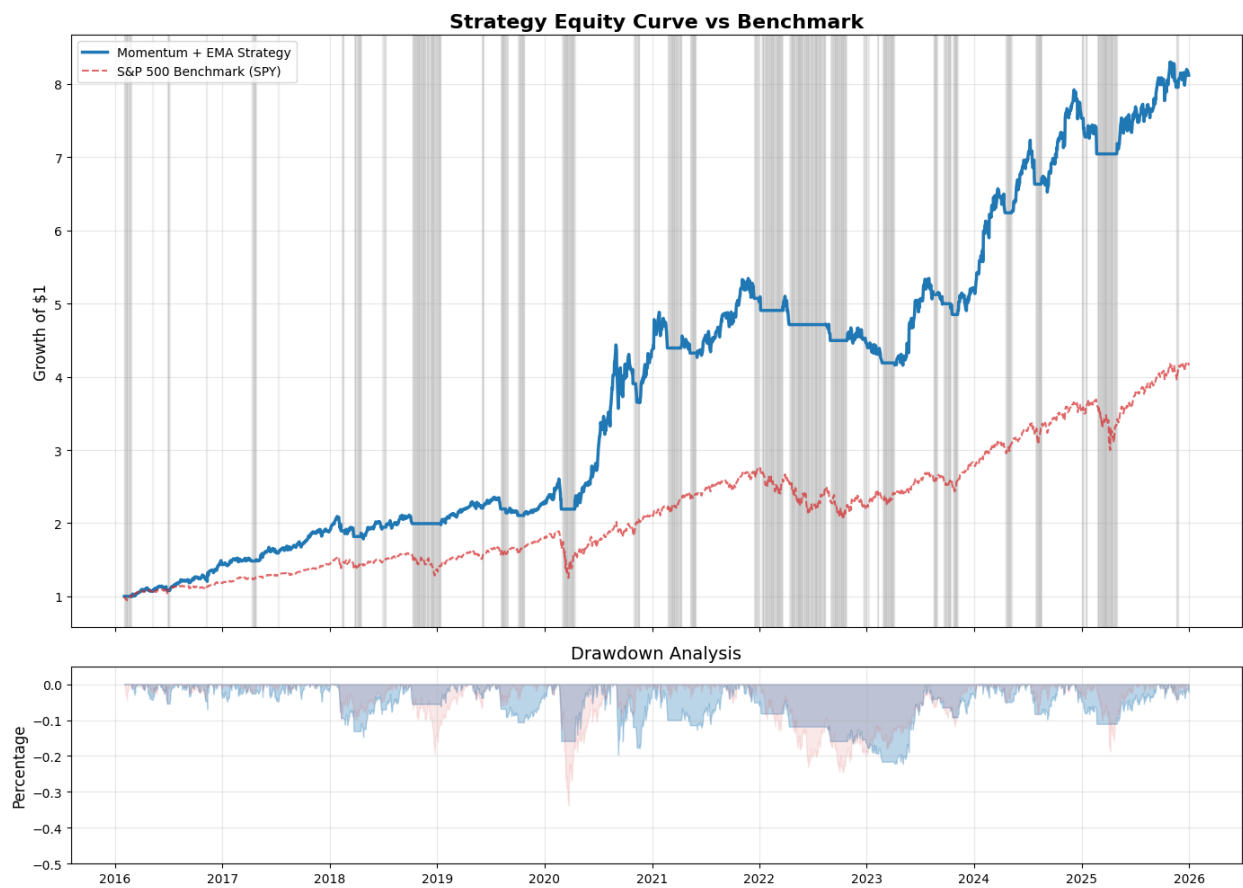


Figure 2. Equity Curve Comparison and Drawdown Duration (2015-2025)

4.2 Performances Measures

To evaluate the performance of the momentum-based trading strategy, several commonly used financial metrics were computed directly from the daily equity curve and return series generated by the backtest (Table 4).

Table 4. Performance Summary (2015 - 2025)

	Strategy	SPY
Total Return	711.90%	315.49%

Annualized (CAGR)	23.93%	15.48%
Volatility	18.15%	17.96%
Sharpe Ratio	1.32	0.86
Max Drawdown	-22.21%	-33.72%

4.2 Yearly Return

Annual returns are analyzed to better understand how the strategy performs across different market regimes. Comparing yearly results with the S&P 500 benchmark helps assess the consistency and persistence of outperformance over time.

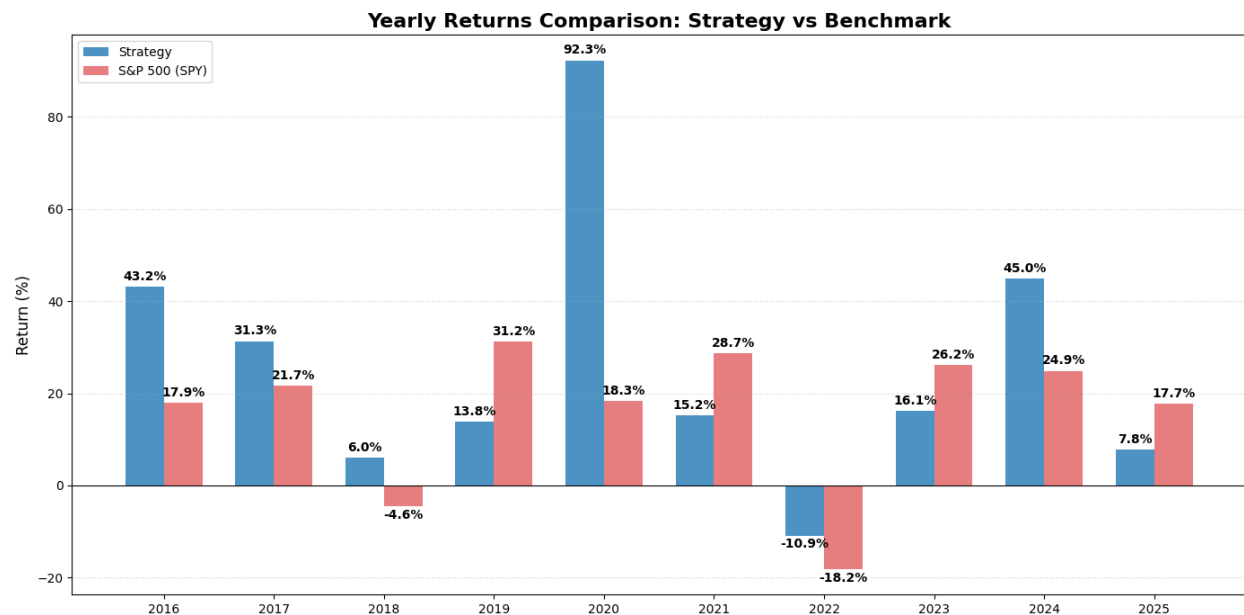


Figure 3. Yearly Returns Comparison

Table 5. Highest year return – 2020

Tickers	Months Held	Average Weight (%)
TSLA	12	29.41%
AAPL	12	12.15%
MSFT	12	8.59%
NVDA	12	13.28%
ADBE	12	6.58%
AMZN	10	8.98%
PYPL	9	7.43%
GOOGL	8	4.33%
META	7	4.22%

UNH	7	3.49%
MA	4	5.74%
HD	4	4.82%
V	3	7.19%
PG	3	5.94%
WMT	3	7.77%
JPM	1	7.16%
DIS	1	2.52%

5. Discussion

5. Limitations

While the results are encouraging, several limitations remain. First, the backtest does not fully account for transaction costs such as slippage and commissions arising from monthly rebalancing, which may reduce realized performance. Second, future research could enhance diversification by incorporating additional factors—such as Value or Quality—alongside Momentum. Finally, further parameter optimization, particularly testing alternative EMA window combinations (e.g., 50/200 days), may improve trend-timing effectiveness across different market regimes.

6. Conclusion

The empirical results show that the proposed momentum-based strategy clearly outperforms the S&P 500 benchmark. From 2015 to 2025, the strategy achieved a compound annual growth rate (CAGR) of 23.93% and a Sharpe Ratio of 1.32, nearly doubling the risk-adjusted performance of a passive buy-and-hold approach.

The combination of inverse-volatility weighting and EMA 9/25 trend filters proved effective in capturing market growth while reducing exposure during major drawdowns, particularly during the market shifts observed in 2020 and 2022.

References

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